AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2019

Computer Science Paper II

Time: 2 hours 15 minutes Marks: 55

April/ May 2019 Teacher for Learning only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are TEN questions. After Question 4, there are TWO sections, section I on C Language and section II on Visual Basic. Attempt EITHER section.
   In each section answer ALL questions. The last question in each section offers TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Forms are preferred to enter data into a database table instead of entering data directly into the table.

Write any THREE reasons to support the given statement. (3 Marks)

b. Define the following database terms. (2 Marks)

- Reports
- Queries

April/ May 2019 for Teaching & Learning only

Q.2. (Total 5 Marks)

a. Ali, a database administrator, wants to apply relationship among currently unrelated tables. The names of these tables are Orders, Items, ItemsOrders, Customers and Addresses.

Suggest the type of relationship that he should apply on the given set of tables.

i. (1 Mark)

Orders			
Order_ID	**Customer_ID**	**Order_Date**	**Amount**
801	101	24/11/2018	14600
802	102	27/11/2018	9800

Items		
Item_ID	**Item_Name**	**Item_Description**
401	Mini Fridge	Eco friendly small-sized fridge.
402	Water Dispenser	Energy efficient, cool and hot water dispenser.
403	Rechargeable Fan	Portable and rechargeable DC fan.

ItemsOrders	
Order_ID	**Item_ID**
801	401
801	402
802	402
802	403

Type of Relationship: _____

April May 2019 for Teaching & Learning Only

ii. (1 Mark)

Customers		
Customer_ID	**Customer_Name**	**Address_ID**
101	Muhammad Ali	501
102	Anusha Mateen	502

Addresses	
Address_ID	**Customer_Address**
501	House No. 50, ABC Block 2, Karachi
502	House No. 80, DEF Block 3, Sukkur

Type of Relationship: _____

iii. (1 Mark)

Customers		
Customer_ID	**Customer_Name**	**Address_ID**
101	Muhammad Ali	501
102	Anusha Mateen	502

Orders			
Order_ID	**Customer_ID**	**Order_Date**	**Amount**
801	101	24/11/2018	14600
802	102	27/11/2018	9800
803	102	30/11/2018	6600
804	101	04/12/2018	3400

Type of Relationship: _____

April ... 2019 ... Teaching & Learning only

b. Read the given database table. (2 Marks)

Student_ID	Student_Name	City
101	Muhammad Ali	Karachi
102	Anusha Mateen	Sukkur
103	Shahid Khan	Sukkur

Which field should NOT be selected as primary key for this database table and why?

April/ May 2019
for
Teaching & Learning only

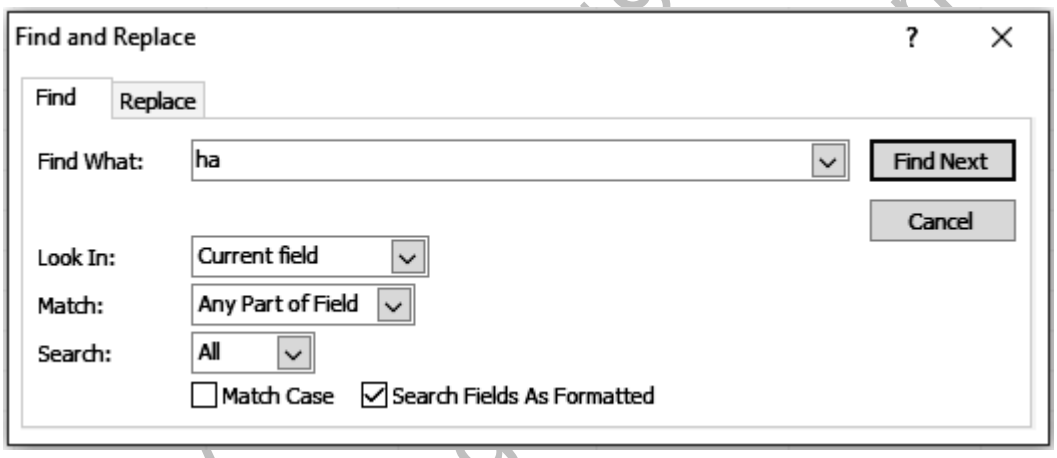
Q.3. (Total 4 Marks)

a. Read the given database table containing the data of employees.

Employee_ID	First_Name	Last_Name	Age
E101	Fahad	Hussain	30
E102	Kashif	Khan	40
E103	Ahad	Zain	50
E104	Hammad	Ali	20

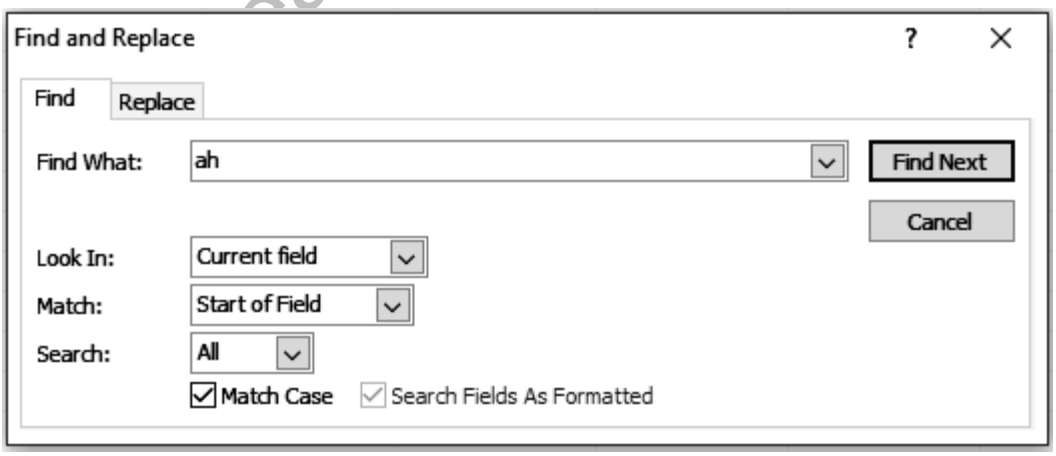
What result would be shown by MS Access in the First_Name field according to the options selected in the given Find and Replace dialogue box?

i. (1 Mark)



Result Shown by MS Access: _____

ii. (1 Mark)



Result Shown by MS Access: _____

b. Read the given database table containing the data of employees.

Employee_ID	First_Name	Last_Name	Age
E101	Fahad	Khan	30
E102	Kashif	Hussain	50
E103	Iqra	Hassan	40
E104	Hammad	Ali	20
E105	Aneela	Omar	35

i. If the ascending sort is applied on **Age** field of this table, then write the **Employee_ID** of record(s) that will appear below E105 in the correct sequence. (1 Mark)

_____

_____

ii. If the descending sort is applied on **First_Name** field of this table, then write the **First_Name** of record(s) that will appear below the Hammad in the correct sequence. (1 Mark)

_____

_____

April/ May 2019 for Teaching & Learning only

Q.4. (Total 6 Marks)

A database table named Hotel_Info is setup to store information about hotels in a city. Few records from the table are shown below.

HID	Name of Hotel	Charges in PKR	Free Breakfast	Over All Rating	Distance from Airport in KM
DL150	Dreamland Hotel	5533	Yes	8	6
BM151	Blue Moon Hotel	3444	Yes	7.8	10
SV152	Sea View Hotel	8565	No	6.3	17
HF153	Hotel Farhan	3699	No	7.5	11
GR154	Green River Hotel	11000	Yes	6.6	5
WP155	White Palace Hotel	4477	No	5.4	18
RB156	Rainbow Hotel	6290	Yes	8.9	3

a. Show the output of the following query in design view of MS Access. (3 Marks)

Field:	HID	Name of Hotel	Charges in PKR	Free Breakfast	Over All Rating
Table:	Hotel_Info	Hotel_Info	Hotel_Info	Hotel_Info	Hotel_Info
Sort:					
Show:	☐	☑	☑	☑	☐
Criteria:			>= 4000	= Yes	
or:					

Space for Output

April/ May/ 2019 for Teaching & Learning only

b. Complete the query in the given design view to select and show the **Name of Hotel** that are not far than a distance of 7 kilometres from the airport or have an **Overall Rating** of above 7.

(3 Marks)

Field:	HID	Name of Hotel	Charges in PKR	Overall Rating	Distance from Airport in KM
Table:	Hotel_Info	Hotel_Info	Hotel_Info	Hotel_Info	Hotel_Info
Sort:					
Show:	☐	☐	☐	☐	☐
Criteria:					
or:					

April/ May 2019
for
Teaching & Learning only

SECTION I: C Language

Q.5. (Total 5 Marks)

The given table shows pairs of characteristics that differentiate between high and low level programming languages.

Identify the characteristic of high level programming languages from each pair by ticking (✓) the relevant box.

(**Note**: One high level programming language characteristic is identified from 1st pair of statements for your assistance.)

S.No.		
1	These programming languages are farther from human languages. ☐	These programming languages are closer to human languages. ☑
2	There is no need to have deep knowledge of hardware to write programs using these programming languages. ☐	Deep knowledge of hardware is required to write programs using these programming languages. ☐
3	These programming languages are used to write application programs. ☐	These programming languages are used to write hardware controlling programs. ☐
4	Programs that are written using these programming languages execute faster. ☐	Programs that are written using these programming languages execute slow. ☐
5	These programming languages provide many built-in functions. ☐	These programming languages do not provide the feature of built-in functions. ☐
6	It is difficult to trace errors in programs that are written using these programming languages. ☐	It is easy to trace errors in programs that are written using these programming languages. ☐

April/ May 2019 for Teaching & Learning only

Q.6. (Total 5 Marks)

a. A Computer Science student writes steps of compilation process. Three of these steps are incorrect and given below. Each of these steps has exactly ONE incorrect word.

Identify the incorrect word in each step by underlining it and write the correct word. (3 Marks)

i. The input given to a compiler is an executable code.

Correct Word: _____

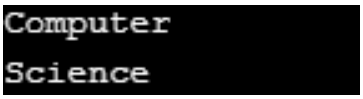
ii. The output of a linker is an object code.

Correct Word: _____

iii. The input of a linker is a source code.

Correct Word: _____

b. Complete the given C program that prints Computer Science as shown in the given image. (2 Marks)



(**Note:** You must use escape sequence in your program to print Science on the next line.)

```
#include <stdio.h>
int main()
{
```

Space for Writing Code

```
 return 0;
}
```

April/ May 2019 for Teaching & Learning only

Q.7. (Total 5 Marks)

Read the given C program.

```
#include <stdio.h>
int main()
{
   int c, n, t, r, x = 0;
   scanf("%d", &n);
   for(c = 1; c <= n; c++)
   {
      x = c * 9;
      r = x % 5;
      t = t + r;
   }
   printf("Result = %d", t);
   return 0;
}
```

a. If the input value in this program is **4** and **for** loop is executed, then complete the given table by writing the missing values of **x**, **r** and **t** for each iteration (repetition) of this program. (4 Marks)

Iteration	x	r	t
1st			
2nd			
3rd			
4th			

b. How many variables are used in this program? (1 Mark)

April/ May 2019 for Teaching & Learning only

Q.8. (Total 6 Marks)

Complete the given code for a user-defined function named 'EvenOdd' that would determine whether an integer is even or odd.

(**Note**: An even number is divisible by 2 while an odd number is not divisible by 2.)

```
#include <stdio.h>
// Declare EvenOdd function here.
[                                                    ]
int main( )
{
   int number;
   printf("Enter an integer: ");
   scanf("%d", &number);
   // Call function here.
   [                                                 ]
   return 0;
}
 // Define function here
[                                                    ]
[                                                    ]
[                                                    ]
```

Q.9. (Total 4 Marks)

Read the given C language code.

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
  int n;
  FILE *fp;
  fp = fopen("C:\\data.txt","w");
  if(fp == NULL)
  {
    printf("Error");
    exit(1);
  }
  printf("Enter number: ");
  scanf("%d",&n);
  fprintf(fp,"%d",n);
  fclose(fp);
  return 0;
}
```

Interpret the purpose of the code lines given from this code.

a. FILE *fp; (1 Mark)

_____

_____

b. fp = fopen("C:\\data.txt","w"); (1 Mark)

_____

_____

c. fprintf(fp,"%d",n); (1 Mark)

_____

_____

d. fclose(fp); (1 Mark)

_____

_____

April/ May 2019 for Teaching & Learning only

Q.10. (Total 10 Marks)

EITHER

a. A gymnasium trainer wants to calculate the maximum, minimum and the average weight of the gymnasium (gym) members.

Write a program using C language that

- inputs weight of 35 gym members.
- calculates minimum, maximum and average weight values.
- outputs minimum, maximum and average weight values.

(**Note**: Use the For loop for repeating the code.)

OR

b. Adil is an owner of a real estate agency. He weekly advertises about the houses for sale in his area. He needs a software to efficiently manage his real estate agency work.

Write a program using C language that:

- inputs the total number of inquiries made by customers on a daily basis in a week.
- inputs the price of properties shared with the customers daily.
- calculates and outputs the number of customers that inquired about the properties worth less than Rs 2,500,000.
- calculates and outputs the percentage of customers that inquired about the properties worth greater than Rs 7,500,000.

April/May 2019 for Teaching & Learning only

April/ May 2019
for
Teaching & Learning only

SECTION II: Visual Basic

Q.5. (Total 5 Marks)

a. Name any TWO properties of forms in Visual Basic. (2 Marks)

b. Describe the following concepts of Visual Basic with the help of an example for each. (3 Marks)

- Objects
- Methods

April/ May 2019 for Teaching & Learning only

Q.6. (Total 5 Marks)

Write a Visual Basic program that

- inputs THREE numeric values via THREE text boxes.
- calculates average of input values.
- sets the size of the characters in the first text box to 14.
- converts the characters in the third text box to boldface.
- outputs average of input values on the label.

April/ May 2019
for
Teaching & Learning only

Q.7. (Total 5 Marks)

a. Differentiate between variables and constants in a programming language. (1 Mark)

b. Describe any ONE use of EACH of the following data types in Visual Basic. (4 Marks)

i. Char

ii. Boolean

iii. String

iv. Single

April/ May 2019
for
Teaching & Learning only

Q.8. (Total 6 Marks)

a. Discuss the purpose of Visual Basic Recordset object with reference to MS Access. (3 Marks)

b. Following are the properties of data control. Describe why a programmer will use these properties while connecting a Visual Basic form with database.

i. Connect (1 Mark)

ii. RecordSource (1 Mark)

iii. DataField (1 Mark)

April/ May 2019 for Teaching & Learning only

Q.9. (Total 4 Marks)

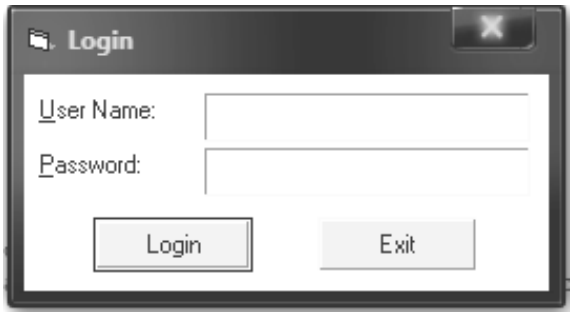
Write a Visual Basic code to draw a black-coloured line with a thickness of 5 pixels. The **x** and **y** coordinates at the starting of line are (0, 0) and ending of line are (500, 1000).

April/ May 2019
for
Teaching & Learning only

Q.10. (Total 10 Marks)

EITHER

a. Write Visual Basic code for the **Login** and **Exit** buttons on a login form as shown. (10 Marks)



The user can login only when the username is **Shahid** and password is **Pakistan**. The following conditions must be met in the code.

- If the correct username and password is entered and **Login** button is pressed, then a message box should appear with message "Login Successful".
- If the username is not entered but the password is entered correctly and **Login** button is pressed, then a message box should appear with message "Username Field Cannot be Left Empty".
- If the incorrect username is entered but the password is entered correctly and **Login** button is pressed, then a message box should appear with message "Incorrect Username".
- If the correct username is entered but the password is not entered and **Login** button is pressed, then a message box should appear with message "Password Field Cannot be Left Empty".
- If the correct username is entered but the password is incorrect and **Login** button is pressed, then a message box should appear with message "Incorrect Password".
- If the Exit button is pressed, then a message box should appear with message "Are You Sure to Exit?" and buttons **Yes** or **No**. If **Yes** is pressed, then login form will be closed. If **No** is pressed, then login form will remain opened.

April/ May 2019 for Teaching & Learning only

Declare Variables:

_____

_____

Code for **Exit** Button:

_____

_____

_____

_____

Code for **Login** Button:

April/ May 2019
for
Teaching & Learning only

OR

b.

i. Differentiate between one dimensional array and two dimensional array with the help of diagrams. (4 Marks)

ii. Write Visual Basic program that (6 Marks)

- inputs 10 integer values in a one dimensional array using input box.
- assigns 0 value to those array elements that are equal to or less than 10.
- outputs the sum of those array elements that are greater than 10 on a form.

(**Note**: Use For loop for input of values in array.)

April/ May 2019
for
Teaching & Learning only

April/ May 2019
for
Teaching & Learning only

END OF PAPER

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only